

SHAHZAD MS

CTO · CIO · CISO · Fractional Tech Executive · Principal Enterprise Architect — Cloud (Azure · AWS · GCP) · AI & Agentic AI · Cybersecurity & Zero Trust ·

DevOps/SRE · Platform Engineering

GTA, Ontario, Canada · PR · On-site, zero notice · shahzad.ms@yahoo.com · linkedin.com/in/shahzadms · shahzadms7.github.io · Winner, Microsoft AI Innovation Hackathon 2025

EXECUTIVE PROFILE

- **Top-1% technologist, 34 years** turning cloud, AI, and cybersecurity investment into measurable enterprise value; **20+ years in architecture and executive leadership**, trusted by boards and C-suites (CTO / CIO / CISO / fractional) across banking, insurance, government & defense, healthcare, retail, legal, and 10+ regulated industries.
- **Multi-cloud at scale** — Azure, AWS, Google Cloud, Oracle, IBM, Alibaba, DigitalOcean: **99.99% uptime** across **5,000+ VMs** and **1,000+ physical servers**, **35% OpEx reduction** and **40% cloud-spend savings** through FinOps and enterprise vendor negotiation (Microsoft EA, AWS EDP, VMware ELA).
- **Production agentic & generative AI** — Claude, OpenAI, Gemini, DeepSeek, Llama, Amazon Nova: model-agnostic orchestration, end-to-end RAG, and multi-agent systems (MCP, A2A), including a **40,000-SKU intelligent recommendation engine** hardened with responsible-AI guardrails and human-in-the-loop governance.
- **Full AI-engineering depth** across Microsoft Foundry, Amazon Bedrock, and Google Gemini Enterprise — MLOps / GenAIOps, model serving (vLLM, Triton, KServe, Ray Serve), vector databases, LLM observability, and AI gateways — turning frontier AI into governed, auditable business capability.
- **AI-native Zero Trust security** — mean time to respond cut **82% (4 hrs → 45 min)**, credential-theft incidents down **92%**, and **zero regulatory or audit findings** across ISO 27001, SOC 2 Type II, and PCI-DSS for **3+ consecutive years**.
- **Enterprise consolidation without business interruption** — **2M+ Active Directory objects** migrated at zero downtime, **10,000+ users** to Microsoft 365 with zero data loss, and **500+ servers** plus legacy email estates modernized to cloud-native — accelerated by Kubernetes, GitOps, IaC, and CI/CD that cut deployment cycles **40%**.
- **Board-level P&L accountability** — **50+ technical staff** and **\$5M+ annual budgets** managed, **\$15M+ revenue enabled**, and **\$2M+ licensing recovered**.
- **Proven global delivery** — **42+ clients across 8 countries and 3 continents**, spanning financial institutions, government & defense, healthcare, insurance, retail, and MSP/MSSP partners.
- **Founder & operator** — Founder / CEO of Alfalah, an AI career-intelligence SaaS serving **195 countries**; Fractional CTO (ENDevo); **Winner, Microsoft AI Innovation Hackathon 2025**.
- **Foundation-to-frontier range** — from legacy datacenter and enterprise systems to cloud-native and AI-native architectures; leads enterprise-wide digital transformation **from strategy through hands-on delivery**.
- **Engagement models** — full-time executive / architect, fractional or interim **CTO · CIO · CISO**, or trusted technical advisor to founders, boards, and portfolio companies.

PROFESSIONAL EXPERIENCE

Principal Cloud, Systems & AI Security Architect

Healthy Planet · Mississauga, ON (On-Site) · Dec 2025 – Jul 2026

Retail enterprise, 45+ locations and 1,200+ employees — full modernization from legacy on-prem datacenters to a secured, AI-enabled Azure hybrid estate.

Enterprise Agentic AI Orchestration Platform (45+ stores). With the business needing **AI adoption without vendor lock-in or security risk**, designed a **model-agnostic orchestration layer** across Claude Opus/Sonnet, OpenAI GPT, Gemini, and **Microsoft Foundry** (Foundry Models, Foundry Agent Service, Foundry Tools, Foundry Control Plane, Foundry IQ). Standardized tool access with **Model Context Protocol (MCP)** and interoperability with **Agent2Agent (A2A)**; integrated Semantic Kernel, Microsoft Agent Framework (MAF), LangGraph, and CrewAI; deployed prompt and hosted agents on **Responses API (Agents v2)** with Agent Registry and Agent Runtime; and enforced comparative **LLM-as-a-judge** evaluation and mandatory human-in-the-loop, issuing Entra Agent IDs via **Agent 365**, codified in Bicep/ARM through Azure DevOps and GitHub Actions with Azure Policy guardrails. **Delivered a governed, production agent platform with a repeatable evaluation harness and zero audit findings.**

RAG Product-Intelligence System (40,000-SKU catalog). To cut support load and lift conversion, engineered a **high-accuracy, low-latency assistant over a 40,000-SKU catalog** using Azure OpenAI and Azure AI Search with custom embeddings and dynamic re-ranking, benchmarking Pinecone, Weaviate, Milvus, Qdrant, Chroma, and LanceDB for cost and accuracy. Cached hot queries in Azure Cache for Redis, held state in Cosmos DB, prepared data in Databricks, served via Azure Functions and API Management, governed metadata in Microsoft Purview, and instrumented quality with MLflow, Langfuse, Arize Phoenix, LangSmith, and Helicone. **Delivered a grounded, cited assistant that reduced support tickets 40%.**

Multi-Modal Azure AI Services Suite. Where teams needed automation across documents, images, voice, and video, delivered a **multi-modal AI layer** with Azure AI Vision (Image Analysis, OCR, Face, Video Indexer, Custom Vision), Azure AI Language (sentiment, NER, PII, summarization, Q&A), Azure AI Document Intelligence, Azure AI Content Safety, Azure AI Anomaly Detector, Azure AI Personalizer, Azure AI Metrics Advisor, Azure AI Health Bot, Azure AI Immersive Reader, Azure AI Bot Service, and Azure AI Translator — powering document classification on Durable Functions, call-center analytics, image/OCR processing, AI Search skillsets, and ML inference on IoT Edge, plus **open-model integration via Hugging Face on Azure** and edge inference with Ollama and llama.cpp, backed by MLOps on Azure Machine Learning, MLflow, Model Registry, Feature Store, and AutoML. **Delivered measurable automation across the store network.**

Speech-to-Speech AI Voice Agent. For safe customer self-service by phone, built an **auditable voice agent** integrating Azure AI Speech (STT/TTS) with Azure Communication Services on an event-driven Azure Functions + API Management flow, layering Azure AI Content Safety, real-time moderation, provenance tracking, human-supervision workflows, and full audit logging via Log Analytics and Microsoft Purview. **Delivered controllable, compliant voice automation.**

Azure Data Platform & SQL Server DBA. To unify analytics, migrated on-prem **SQL Server (4 TB, 3.1-billion-row)** to Azure SQL Managed Instance with deep query/index tuning, then architected **Bronze-Silver-Gold medallion** layers on Microsoft Fabric / OneLake ingested via Azure Data Factory, Synapse, HDInsight, Stream Analytics, and Databricks, serving Power BI and Zoho Analytics, and stood up managed PostgreSQL, MySQL, Cassandra, DocumentDB, Analysis Services, Data Explorer, Event Hubs, Data Share, Time Series Insights, and Microsoft Graph Data Connect. **Delivered unified, revenue-enabling analytics across 45+ locations.**

Zero Trust SOC, Identity, IoT/OT & Distributed Network (45+ sites). Facing a distributed retail attack surface, deployed **Zero Trust** with Microsoft Entra ID, Conditional Access, MFA, Entra ID Governance, Entra Suite, PIM/JIT, and Intune, on an Azure Virtual WAN hub-and-spoke (ExpressRoute, VPN Gateway, Azure Firewall, WAF, DDoS Protection, Bastion, Private Link, Front Door, CDN), segmenting robotics/IoT/OT behind Azure IoT Hub, IoT Edge, IoT Central, IoT Operations, DPS, Microsoft Defender

for IoT, and Azure Digital Twins, and centralizing detection in Microsoft Sentinel, Defender XDR, Defender for Cloud, Defender EASM, and Security Copilot aligned to MITRE ATT&CK/D3FEND. **Delivered unified threat visibility and Zero-Trust access across 45+ sites.**

Enterprise Hybrid Datacenter, Edge & Disaster Recovery. To guarantee resilience and real-time store automation, architected **three geo-redundant Azure environments** (Canada Central primary, Canada East DR), deployed Azure Local (Azure Stack HCI) on HPE/Dell/Lenovo hyperconverged hardware unified via Azure Arc, engineered a **sub-10ms edge** driving 20+ in-store robotic units (**Physical AI**), and built redundant ESXi/Hyper-V with Azure Site Recovery and Azure Backup, auto-scaling on VM Scale Sets, Spot VMs, Container Apps, and Functions over Blob, Data Lake, Files, NetApp Files, and Elastic SAN. **Delivered guaranteed RPO/RTO SLAs with automated failover.** Also modernized ERP from Dynamics NAV 2017 to **Dynamics 365 Business Central** with zero disruption, and provisioned the core Azure backbone (Virtual Network, DNS, Traffic Manager, ACR/ACI, Logic Apps, Service Bus, Event Grid, Web PubSub, SignalR, Key Vault/Managed HSM, Chaos Studio, App Service, AVS).

Enterprise Knowledge & Document-Intelligence Platform. To let 45+ stores run on trusted AI, **composed a single knowledge platform** on a hardened Foundry landing zone: baseline and basic Foundry chat with AI-agents-at-scale (agent selector / factory pattern, semantic cache) and multi-agent workflow automation, all behind an Azure OpenAI gateway with monitoring and custom Entra auth. Wired in document classification on Durable Functions, automated PDF-forms processing, extract-object-text OCR, search-blob-metadata, AI Search skillsets, generate-documents-from-data, and unlock-insights-from-conversational-data; trained and shipped models via build-and-deploy custom models and many-models with Azure ML; added video analysis with computer vision & ML, intelligent image-processing apps, and open models through Hugging Face on Azure with Ollama and llama.cpp edge inference; and isolated sensitive work on secure compute for research (Confidential Computing / TEE). **Delivered MVP → pilot → enterprise with 40% ticket reduction and zero audit findings.**

Retail Data, Forecasting & Real-Time Analytics Platform. With leaders needing **one source of truth**, engineered an end-to-end data platform combining a greenfield lakehouse with Fabric / OneLake and enterprise BI, a small-medium and modern data warehouse on Synapse, stream processing with Databricks and a real-time lakehouse, data streaming on Event Hubs + Stream Analytics, ETL / ELT with DataOps (MDW), ingest-ETL-stream with Azure Databricks, Sync MongoDB Atlas to Fabric, and IoT analytics on Azure Data Explorer — then layered next-order forecasting, orchestrate-ML and modern analytics with Databricks, real-time and batch scoring, predictive maintenance, and anomaly detection. **Delivered unified analytics and demand forecasting across 45+ locations.**

Edge, Hybrid & Physical-AI Store Automation. To automate the physical store, unified Azure Local baseline and switchless clusters with AKS Hybrid on Azure Local and Azure Virtual Desktop on Azure Local (running ESRI ArcGIS on AVD), drove robotics through ML inference on IoT Edge, Industrial IoT, and IoT private file upload as Physical AI, and added quantum computing integration with classical apps plus app-sustainability measurement (SCI score). **Delivered sub-10ms edge automation with measured carbon efficiency.**

Fractional CTO

ENDevo • Remote • Jan 2026 – Present

Board-level technical leadership on cloud adoption, AI transformation, and security modernization for a legacy-readiness SaaS platform serving North American enterprises.

Cloud-Native Tier-4 AWS SaaS Platform. With the board needing a **scalable, resilient production foundation**, architected an end-to-end Tier-4 SaaS on AWS — EC2 Auto Scaling, EKS/Fargate, RDS Aurora Multi-AZ, S3, CloudFront, API Gateway, Lambda, Cognito, VPC, Transit Gateway, Direct Connect, WAF, Shield Advanced, GuardDuty, Security Hub, KMS, Secrets Manager, Control Tower, Organizations, ECR — with microservices, Istio/Linkerd service mesh, Terraform and AWS CDK IaC, CodePipeline/GitHub Actions CI/CD, and blue-green/canary releases observed via CloudWatch, X-Ray, Datadog, and CloudHealth. **Delivered 99.99% uptime and 40% infrastructure cost reduction.**

Amazon Bedrock Generative-AI Guidance. To differentiate the product, integrated **Amazon Bedrock** with Claude Sonnet 4.6 / Opus 4.6 and **Amazon Nova**, enforcing tenant isolation via **Bedrock AgentCore** across documents, knowledge base, memory, and cost tracking, with Bedrock Knowledge Bases, Guardrails, and Evaluation, plus SageMaker MLOps (Pipelines, Model Registry, Feature Store), Amazon Q, Kendra, Textract, and Comprehend. **Delivered governed, differentiated AI features aligned to product goals.**

Anthropic Claude Enterprise Agentic Suite (multi-cloud). For safe, portable agent operations, deployed **Claude Managed Agents** (sandboxed, MCP-connected), **Claude Code** for agentic coding, **Claude Cowork** for desktop collaboration, **Claude Science** for research, **Voice Mode**, and **Claude Skills**, standardized on MCP and A2A and run vendor-agnostically across Amazon Bedrock, Google Vertex AI, and Microsoft Foundry. **Delivered governed, vendor-agnostic agentic AI.**

Cross-Cloud Identity, Governance & Agent Operations. For safe multi-cloud operations, ran **AI-agents-at-scale** behind a governed model gateway (monitoring, custom auth, semantic cache), integrated **AWS with Azure AD (Entra) security**, orchestrated OpCon-Azure jobs, and executed migrate-cloud-workloads-across-security-tenants with strict multitenant isolation. **Delivered governed, portable operations spanning AWS and Azure.**

Founder & CEO

Alfalah • Remote • 2023 – Present

AI career-intelligence SaaS serving 195 countries with 19 insight modules per report.

AI Career-Intelligence Platform (19 modules, 195 countries). Seeing job-seekers priced out by **\$49–149/month incumbents**, built a low-cost, privacy-first platform: launched the **~\$20/month MVP** on Next.js, Supabase (PostgreSQL + pgvector), Vercel, Clerk, Stripe, Cloudflare, DeepSeek V4 Flash, Pinecone, Upstash Redis, PostHog, Sentry, Resend, and GitHub; piloted **Haystack multi-model routing** (DeepSeek V4 Pro, GLM-5.2, Kimi K2.7 Code, Gemma 4) with Qdrant hybrid retrieval, PostgreSQL row-level security, and prompt caching plus LLM-as-a-judge; and scaled globally with Cloudflare Workers, AWS Lambda, and Stripe Connect. **Delivered a differentiated product across 195 countries at a 50x cost reduction.**

Open-Source RAG & Multi-Agent Platform. To stay vendor-neutral and self-hostable, engineered an enterprise open-source stack — multi-agent orchestration (LangChain, LangGraph, LlamaIndex, Haystack, Semantic Kernel, AutoGen, CrewAI, DSPy, OpenAI Agents SDK, Google ADK, Mastra); a poly-store vector layer (Pinecone, Weaviate, Milvus, Qdrant, Chroma, LanceDB, Vespa, pgvector, Redis Vector, MongoDB Atlas Vector Search, FAISS, TiDB); high-throughput serving (vLLM, Triton, KServe, Ray Serve, Ollama, llama.cpp, TensorRT-LLM, TGI, SGLang, Together AI, Fireworks AI, Modal); full MLOps (MLflow, Kubeflow, Weights & Biases, DVC, ClearML, BentoML, ZenML, Metaflow, Flyte, TrueFoundry); observability (Langfuse, Arize Phoenix, LangSmith, Helicone, Weave, Opik, Braintrust, SigNoz, OpenTelemetry); AI gateways (LiteLLM, Portkey, OpenRouter, Cloudflare AI Gateway, Nexus AI Gateway); and enterprise-AI platforms (Databricks, Dataiku, DataRobot, H2O.ai, Domino Data Lab, Alteryx). **Delivered a self-hostable, ~99.9%-uptime AI platform.** Engineered privacy by design with row-level security, one-click data deletion, MFA, and a zero-data-sold policy.

Founder & Principal Consultant | Fractional Technology Advisor

Independent Practice (Self-Employed) • Greater Toronto Area • 2023 – Present

Fractional advisory and hands-on delivery for founders, co-founders, and CTOs across web and SaaS.

Fractional CTO & SaaS Delivery. Where founders had **technical gaps from strategy to build**, translated business goals into HLD/LLD, roadmaps, and TOGAF reference architectures, then built secure-by-design SaaS with cloud-native compute, managed databases, authentication, APIs, CI/CD, and AI/LLM integration, and grew reach through responsive web design, SEO, and Google Ads (Search, Display, Video, Shopping) with Google Analytics. **Delivered end-to-end products and measurable client growth.**

Senior Solution Architect | L3+ Systems & Cloud Engineer | SRE Specialist

CCS • Toronto, ON • 2021 – 2025

Mission-critical infrastructure modernization across 5,000+ VMs and 1,000+ physical servers.

Multi-Cloud Landing Zones & AI-Ready Infrastructure. Needing **governed, cost-optimized foundations** across five clouds, architected landing zones on Azure (VNet, ExpressRoute, Firewall, App Gateway, Front Door, Bastion, Private Link), AWS (EC2, multi-AZ VPC, Direct Connect, Transit Gateway, RDS/Aurora, ECS/EKS, GuardDuty, Control Tower), and GCP (Compute Engine, GKE, Cloud SQL, Cloud Armor, Vertex AI, BigQuery, Shared VPC), with GPU-accelerated compute for AI/ML and FinOps (reserved instances, savings plans, spot). **Delivered 99.99% uptime across five clouds.**

IaC, CI/CD, Kubernetes & Platform Engineering. To eliminate drift and slow releases, built **10K+ lines of Terraform** plus 500+ Ansible playbooks, Pulumi, CloudFormation, ARM, and Bicep across GitHub Actions, GitLab CI/CD, and Azure Pipelines with Policy-as-Code (Azure Policy, AWS Config, OPA), deployed multi-cloud Kubernetes (EKS, GKE, AKS, Arc) with Helm, Istio, and Linkerd, and operationalized GitOps with ArgoCD and Flux — plus internal developer platforms (Backstage, Port, Humanitec, Cortex, Crossplane) and chaos engineering (Gremlin, Chaos Toolkit, Litmus, Azure Chaos Studio, Chaos Mesh). **Delivered 40% faster, drift-free deployments.**

Early GenAI / MLOps Enablement. As clients began adopting GenAI, stood up the **AI-ready foundation** — Azure Machine Learning, MLflow, Kubeflow, Weights & Biases, DVC, ClearML, Feature Store, Model Registry, and AutoML, GPU model serving (Triton, KServe, Ray Serve, vLLM), a benchmarked vector layer (Pinecone, Weaviate, Milvus, Qdrant, Chroma, FAISS, pgvector, Redis Vector), a Databricks/Delta Lake/Unity Catalog lakehouse, and LLM observability (Langfuse, Arize Phoenix, LangSmith) behind a LiteLLM gateway. **Delivered a reusable AI/MLOps platform enabling client GenAI adoption.**

Zero Trust, Identity, SIEM/XDR & Security Operations. To shrink risk across the estate, deployed Entra ID Conditional Access, MFA, PIM/JIT, Windows Hello, FIDO2, and SAML/OAuth/OIDC with CyberArk, BeyondTrust, and Delinea (federating 50+ apps), and Microsoft Sentinel, Defender XDR, CrowdStrike Falcon, SentinelOne, Darktrace, Netskope, and Zscaler on a CNAPP strategy (CSPM, CIEM) aligned to MITRE ATT&CK/D3FEND. **Delivered MTTR down 82%, password incidents down 92%, and alert fatigue down 70%.**

Migrations, BCDR, Endpoints & Compliance. Consolidating without downtime, migrated **2M+ Active Directory objects** (ADMT, Quest) and **10,000+ users / 500+ servers** at petabyte scale (Azure Migrate/ASR, AWS MGN), engineered immutable BCDR (Veeam WORM, Azure Site Recovery, Acronis, Rubrik, Commvault, Zerto, Datto), managed **5,000+ endpoints** (Intune, SCCM/MECM, Jamf, Autopilot) at 98% compliance, and sustained ISO 27001, NIST CSF, SOC 2 Type II, PCI-DSS, and HIPAA/HITECH with Drata/Vanta and HashiCorp Vault. **Delivered 35% OpEx reduction and zero audit findings for 3+ years.**

Observability, AIOps & SRE Transformation. To improve reliability and cut toil, deployed Datadog, Dynatrace, Prometheus/Grafana (500+ dashboards), Azure Monitor/Log Analytics, and ELK (50 TB+/day), built an **Azure OpenAI SRE copilot** for automated incident response, and led SRE transformation with SLO/SLI/SLA, error budgets, and chaos engineering while managing **50+ staff and a \$5M+ budget** and negotiating Microsoft EA, AWS EDP, and VMware ELA. **Delivered reliability from 98.5% to 99.99%, \$15M+ revenue enabled, and \$2M+ licensing savings.**

Enterprise Container & Microservices Platform. Standardizing delivery, built an AKS platform spanning baseline, mission-critical, and multi-cluster / multi-region topologies with AKS microservices (advanced) behind Front Door, microservices with Dapr, and Azure Red Hat OpenShift for financial-services workloads, shipped through blue-green and canary for AKS, a GitOps blueprint, and CI/CD for microservices. **Delivered portable, policy-governed workloads at four-nines.**

Cloud-Native Resilience & Integration Patterns. To harden reliability, engineered event-driven systems on Event Hubs + Functions with serverless observability, composing resilient design patterns end-to-end — CQRS, event sourcing, saga, claim-check, cache-aside, materialized view, index table, gateway routing / offloading / aggregation, valet key, pipes-and-filters, messaging bridge, scheduler-agent-supervisor, circuit breaker, bulkhead, sidecar, ambassador, anti-corruption layer, strangler fig, retry, priority queue, competing consumers, leader election, quarantine, health-endpoint monitoring, static content hosting, and externalized configuration. **Delivered fault-tolerant, self-healing services.**

Defense-in-Depth Security Operations. Reducing risk across the estate, delivered map-threats (Sentinel + MITRE ATT&CK), Sentinel automated response, Microsoft 365 Defender integration, and first- and second-layer defense with multi-layered VM protection at massive scale, plus secure SQL MI with Managed HSM, forensics, certificate lifecycle, VM compliance, noisy-neighbor mitigation, and a performance & security-optimized Virtual WAN. **Delivered MTTR down 82% with zero audit findings.**

Multi-Cloud & Hybrid Networking. Connecting five clouds, architected hub-spoke and Virtual WAN with hybrid and split-brain DNS, Azure DNS Private Resolver, Private Link with Virtual WAN DNS (single-region and virtual-hub extension), IPv6, ExpressRoute / VPN failover and troubleshooting, virtual network peering, trusted internet connections, Azure Arc SQL Server, hybrid performance monitoring, and Azure File Share. **Delivered secure, resilient global connectivity.**

Modern Data & MLOps Platform. Enabling analytics and AI, built a greenfield lakehouse and enterprise BI on Fabric, a small-medium and modern data warehouse on Synapse, an end-to-end data platform, stream processing and a real-time lakehouse on Databricks, data streaming with Stream Analytics, ETL / ELT with DataOps (MDW), ingest-ETL-stream with Databricks, combine-relational-and-NoSQL stores, and IoT with Azure Data Explorer, operationalized with MLOps for Python, distributed training, and real-time / batch scoring. **Delivered a reusable data + MLOps foundation.**

DevSecOps & Platform Engineering. Accelerating teams, delivered DevSecOps Infrastructure-as-Code, Automated API Deployments (APIOps), GitOps, canary and feature flags, Policy-as-Code, platform engineering and DevEx, chaos engineering, and managed M365 tenant configuration (Microsoft365DSC + DevOps) with serverless observability. **Delivered 40% faster, drift-free, self-service delivery.**

Senior Systems Architect | L3+ Cloud & Infrastructure Engineer

Lewis and Associates LLP • Toronto, ON • 2020

Hybrid multi-cloud, virtualization, security, and Microsoft 365 engineering for legal and financial-services clients.

Hybrid Multi-Cloud, Security & M365 Modernization. With legal/financial clients needing **resilient, compliant infrastructure**, designed a hybrid estate across Azure, AWS, GCP, and on-prem unifying 500+ VMs — VMware vSphere 6.7/7.0 (vSAN, HA/DRS), Hyper-V (SCVMM, Storage Spaces Direct), and NSX-T microsegmentation at 99.9% uptime — automated with Terraform (20K+ lines), Ansible, and PowerShell DSC under CIS-benchmark Policy-as-Code, protected 500+ users with Entra ID and Sentinel/Defender/Splunk ES/CrowdStrike under NIST SP 800-53, and migrated 2,000+ mailboxes to Exchange Online and 500 TB+ to SharePoint Online (BitTitan, Quest, SkyKick, CloudM). **Delivered zero-compromise, zero-data-loss modernization.**

Secure Legal Application, API & Web Modernization. To modernize client-facing systems, delivered fully managed secure apps and API protection (Azure App Service, App Service Environment, API Management, Application Gateway, Entra ID, Key Vault, Front Door, WAF), a highly available SharePoint farm, zone-redundant and DR-ready multi-tier apps, private-resolver and split-brain DNS, and ExpressRoute/VPN failover, run under ITIL with ServiceNow, Jira, and Confluence. **Delivered secure, highly available legal workloads.**

Secure App, API & Web Estate. Modernizing client systems, delivered fully managed secure apps and protect-APIs with an APIM API scenario, App Service Environment (HA and standard), basic and zone-redundant web apps, N-tier Windows and Linux VM apps, a highly available SharePoint farm, WordPress on App Service and container, multi-tier app disaster recovery, and scalable-apps performance modeling under SDLC with DevOps and APIOps. **Delivered secure, highly available legal and financial workloads.**

Hybrid Messaging & Connectivity. Bridging on-prem and cloud, delivered secure hybrid messaging (mobile / web / client), Azure Arc SQL Server, Azure File Share (hybrid and private), hybrid performance monitoring, hybrid DNS, ExpressRoute / VPN failover, trusted internet connections, multitenant considerations, extend-on-prem-AD-domain, SQL Managed Instance disaster recovery, certificate lifecycle, and VM compliance. **Delivered compliant, resilient hybrid connectivity.**

Senior Cloud Architect | L3+ Infrastructure Engineer (progressed from Systems Administrator)

Cronomagic Canada Inc. • Montreal, QC • 2000 – 2019

20-year progression across enterprise and MSP/MSSP clients in carrier-neutral colocation serving Canada's top ISPs — the legacy backbone of a 34-year career.

Foundational Systems & Datacenter Operations. Running **carrier-neutral colocation** for top ISPs, administered the full physical datacenter lifecycle (rack/cage, structured cabling, A/B power, PDU/UPS, CRAC cooling, cross-connects, carrier handoffs) and deployed Windows Server 2000/2003/2008 and Red Hat Linux, Active Directory (all five FSMO roles), Exchange 2000/2003/2007, IIS, DNS/DHCP/WINS, Cisco routing/switching, early VMware ESX and Hyper-V on EMC Clariion/NetApp/HP SAN, and SQL Server with failover clustering, developing apps in C#/ASP.NET and monitoring with MRTG/Cacti/Nagios. **Delivered reliable, carrier-grade enterprise and MSP infrastructure.**

Enterprise Cloud Transformation & Modernization (2009–2019). Leading a **10-year transformation**, migrated 5,000+ VMs and 1,000+ physical servers to a cloud-first, hybrid, edge-aware model — HCI (Nutanix, vSAN, Storage Spaces Direct), vSphere 6.x/7.0, Hyper-V SCVMM, RHEV, IaC (Terraform, Ansible, Puppet, Chef), CI/CD (Azure DevOps, GitHub Actions, GitLab), SDN (NSX-T), and multi-region BCDR (Azure Site Recovery, AWS Elastic Disaster Recovery, Zerto, Veeam), with observability (SolarWinds, PRTG, Datadog, Dynatrace, Prometheus/Grafana, ELK) and AIOps alerting. **Delivered RPO < 1 hr, RTO < 2 hrs, and 30% cost reduction.**

Carrier-Grade Mail, DNS, Web & Messaging (MSP/MSSP). For ISP and MSP/MSSP clients, built and operated **multi-tenant messaging and hosting at carrier scale** — Postfix, Sendmail, Exim, qmail, Dovecot, and Cyrus with SpamAssassin, ClamAV, Amavis, Rspamd, OpenDKIM, and OpenDMARC; commercial platforms (Zimbra, Xigen, MDAemon, Kerio, IceWarp, IBM Domino, GroupWise); webmail (Roundcube, SquirrelMail, RainLoop, Horde); authoritative/recursive DNS (BIND, PowerDNS, Unbound, Knot, NSD, djbdns); and web/app stacks (Apache, NGINX, Tomcat, JBoss, WebLogic, WebSphere) over MySQL, PostgreSQL, and Oracle. **Delivered secure, reliable hosting and messaging for Canada's top ISPs.**

Messaging Migration, HA & Legacy Modernization. Executed **zero-data-loss migrations** of 10,000+ Exchange mailboxes and legacy MTA estates (Postfix, Sendmail, Exim, qmail, Zimbra, Xigen, MDAemon) to Exchange Online and 2,000+ SharePoint sites (500 TB+) via BitTitan/Quest/SkyKick/CloudM/SPMT, engineered HA (SQL Server Always-On, Exchange DAG, Hyper-V Replica, vSphere HA/DRS, WSFC) with F5 BIG-IP/NGINX/HAProxy, and modernized legacy applications and databases (COBOL, FORTRAN, VB, Delphi, PowerBuilder, FoxPro, Clipper, ColdFusion; Oracle RAC, Sybase, Ingres, Gupta SQLBase, InterBase, Pervasive) across UNIX/midrange estates (AIX, Solaris, HP-UX, SCO, NetWare, OS/2, OS/400, VMS). **Delivered continuity while paving a cloud path**, and negotiated a Microsoft EA saving \$2M+.

Mainframe & Midrange Modernization. Moving legacy estates to Azure, delivered Unisys ClearPath Forward rehost and OS 2200 virtualization, IBM iSeries and AIX to Azure (Skytap), AIX to Azure Linux, general mainframe refactor and re-engineering, process-batch-transactions, rehost IMS with Raincode, refactor Adabas to AKS and rehost Adabas (Software AG), integrate IBM message queues (IBM MQ), extend mainframes with REST APIs, mainframe data replication (Data Factory / Qlik / Precisely), and move / archive mainframe data. **Delivered continuity with a clear cloud path.**

Oracle, SAP & Enterprise Applications on Azure. For enterprise workloads, delivered Oracle to Azure IaaS, Oracle Exadata (ODAA), migrate-Oracle-to-Azure-SQL, SAP HANA, SAP BW/4HANA, SAP production, IBM Maximo, Teamcenter PLM on NetApp Files, and Moodle on NetApp Files. **Delivered mission-critical apps at guaranteed performance.**

Web, N-Tier & Highly Available Hosting. For ISP and MSP clients, delivered N-tier Windows and Linux VM apps, basic and zone-redundant web apps, WordPress hosting, a highly available SharePoint farm, multi-region web apps, an e-commerce front end with scalable order processing, hub-spoke and hybrid network reference, hybrid DNS, Azure File Share and Files (private), virtual network peering, enterprise file-share DR, and multi-region BCDR. **Delivered reliable, scalable hosting at carrier grade.**

FLAGSHIP SOLUTION PORTFOLIO

Enterprise-scale reference builds carried from proof-of-concept to enterprise delivery, demonstrating maximum-combination solution delivery across each platform ecosystem.

Azure — Microsoft Foundry Enterprise AI & Data Platform. A complete, governed Azure AI operating platform: baseline Foundry Chat with grounding and Content Safety, a Foundry Landing Zone with governance and cost controls, then enterprise multi-agent workflow automation, AI-agents-at-scale on Foundry Agent Service and AKS, document classification (Durable Functions + Document Intelligence), call-center OpenAI analytics, generate-documents-from-data, multi-modal content processing (Vision, Language, Speech, Video Indexer), AI Search skillsets, build-and-deploy custom models, OpenAI gateway monitoring with custom Entra ID auth, ML inference on IoT Edge, secure compute for research (Confidential VMs/TEE, Managed HSM), and quantum integration with classical apps (Azure Quantum, Qiskit, Cirq) — on Fabric/OneLake/Databricks with MLflow, Model Registry, Feature Store, and AutoML. **Covers every Foundry and Azure AI service end to end.**

AWS — Tier-4 Generative-AI SaaS on Bedrock + SageMaker. Amazon Bedrock (AgentCore, Knowledge Bases, Guardrails, Evaluation) with Claude and Amazon Nova, SageMaker AI & Studio with SageMaker MLOps (Pipelines, Model Registry, Feature Store, Ground Truth), Amazon Q & Q Developer, OpenSearch vector RAG, Kendra enterprise search, Textract with Comprehend & Comprehend Medical document AI, CodeGuru, HealthLake, DevOps Guru, Rekognition/Polly/Transcribe/Lex, Personalize/Forecast/Fraud Detector, and Redshift/Glue/Athena/EMR/Kinesis/MSK/QuickSight analytics on EC2/EKS/Fargate/Lambda, delivered via CDK/CodePipeline with blue-green/canary and full observability. **99.99% uptime, 40% cost reduction; every core AWS AI/ML service exercised.**

GCP — Gemini Enterprise Agent Platform. Gemini Enterprise Agent Platform (Agent Studio, Agent Registry, Agent Runtime, Memory Bank, Agent Simulation, Google ADK) with Gemini 3.1 / Gemma 4, Lyria 3 and Imagen, Model Garden, Vertex AI (Vector Search, Pipelines, Workbench, AutoML, Explainable AI), BigQuery ML, Dataflow, Dataproc, Dataplex, Looker, Cloud Spanner and AlloyDB analytics, Document AI, Vision AI, Video AI, Speech-to-Text, Text-to-Speech, Translation AI, Natural Language AI,

Recommendation AI, and Contact Center AI + Dialogflow CX on Compute Engine, GKE, Cloud Run, Cloud Functions and Cloud TPU, secured by Chronicle SIEM, Security Command Center, and BeyondCorp, delivered via Apigee/Cloud Build/Anthos with Terraform. **Full-lifecycle Gemini Enterprise agent platform at 99.99% uptime.**

Anthropic Claude — Enterprise Agentic AI Suite. Claude Opus 5, Sonnet 4.6, Haiku, Claude Code, Claude Managed Agents, Claude Cowork, Claude Science, Voice Mode, and Claude Skills, standardized on MCP and A2A with subagents and agent skills, orchestrated via LangGraph/CrewAI behind a LiteLLM gateway with Langfuse/Arize Phoenix observability, and run vendor-agnostically across Bedrock, Vertex AI, and Foundry. **Governed, portable, enterprise-grade Claude operations.**

Open-Source — Sovereign RAG, Agent & MLOps Platform. A vendor-neutral, self-hostable platform: multi-agent orchestration (LangChain, LangGraph, LlamaIndex, Haystack, CrewAI, AutoGen, DSPy, MAF, OpenAI Agents SDK, Google ADK, Mastra), poly-store vector search, a full serving fleet (vLLM, Triton, KServe, Ray Serve, Ollama, llama.cpp, TensorRT-LLM, TGI, SGLang, Together AI, Fireworks AI, Modal), complete MLOps (MLflow, Kubeflow, W&B, DVC, ClearML, BentoML, ZenML, Metaflow, Flyte, TrueFoundry), full observability and evaluation, all AI gateways, and enterprise-AI platforms (Databricks, Dataiku, DataRobot, H2O.ai, Domino, Alteryx) on Kubernetes. **Covers the full open-source AI stack.**

SKILLS, PLATFORMS & TECHNOLOGY STACK

Cloud & Multi-Cloud

Platforms: Microsoft Azure, AWS, Google Cloud Platform, Oracle Cloud (OCI), IBM Cloud, Alibaba Cloud, DigitalOcean, Vultr, Heroku; Hybrid & cloud-native architecture, Azure Landing Zones, Cloud Adoption & Well-Architected Frameworks, Azure Arc, Azure Migrate, AWS Migration Hub, GCP Migrate; Cloud governance, FinOps & cost optimization, multi-cloud strategy, Virtual WAN, hub-spoke, management groups, Shared VPC, Workload Identity, Private Service Connect.

AWS AI & cloud: Amazon Bedrock (AgentCore, Knowledge Bases, Guardrails, Evaluation), Amazon Nova, Amazon Q & Q Developer, SageMaker AI (Studio, Pipelines, Model Registry, Feature Store, Ground Truth, JumpStart), Rekognition, Polly, Transcribe, Lex, Comprehend & Comprehend Medical, Texttract, Translate, Kendra, Personalize, Forecast, Fraud Detector, CodeGuru, HealthLake, DevOps Guru; EC2, EKS / Fargate, Lambda, S3, RDS / Aurora, DynamoDB, Redshift, Glue, Athena, EMR, Kinesis, MSK, OpenSearch, API Gateway, Step Functions, Control Tower, GuardDuty, Security Hub, KMS, CloudFormation, CDK, CodePipeline.

Google Cloud AI & core: Gemini Enterprise Agent Platform (Agent Studio, Agent Registry, Agent Runtime, Memory Bank, Agent Simulation), Google ADK, Gemini 3.1 Pro / Flash, Gemma 4, Lyria 3, Imagen, Model Garden, Vertex AI (Vector Search, Pipelines, Workbench, AutoML, Custom Training, Prediction, Explainable AI), Document AI, Vision AI, Video AI, Speech-to-Text, Text-to-Speech, Translation AI, Natural Language AI, Dialogflow CX, Contact Center AI, Recommendation AI, Cloud TPU; Compute Engine, GKE, Cloud Run, Cloud Functions, BigQuery & BigQuery ML, Dataflow, Dataproc, Dataplex, Looker, Pub/Sub, Cloud Spanner, AlloyDB, Firestore, Bigtable; Chronicle SIEM, Security Command Center, BeyondCorp Enterprise, Apigee, Anthos, Cloud Build.

AI, Agentic AI & LLM Engineering

Models: Anthropic Claude (Opus 5, Opus 4.6, Sonnet 4.6, Haiku, Claude Code, Managed Agents, Cowork, Science, Platform, Voice Mode, Skills), OpenAI GPT (GPT-5.x, GPT-4.x, o-series, GPT-4o audio, gpt-image-2), Google Gemini (3.1 Pro/Flash, Gemma 4), DeepSeek V4 Flash/Pro, GLM-5.2, Kimi K2.7 Code, Phi, Amazon Nova (Micro, Premier, Canvas, Reel), Llama, Mistral, Cohere.

Microsoft Foundry & Azure AI: Foundry Models, Foundry Agent Service (Prompt & Hosted Agents), Foundry Tools, Foundry Control Plane, Foundry IQ, Agent 365, Responses API (Agents v2), Agent Registry, Agent Runtime; Azure OpenAI, Azure AI Search, Azure Machine Learning; Azure AI Vision (Image Analysis, OCR, Spatial Analysis, Face, Video Indexer, Custom Vision), Azure AI Language (sentiment, NER, PII, summarization, Q&A), Azure AI Speech (STT/TTS/translation), Azure AI Document Intelligence, Azure AI Content Safety, Azure AI Anomaly Detector, Azure AI Personalizer, Azure AI Metrics Advisor, Azure AI Health Bot, Azure AI Immersive Reader, Azure AI Bot Service, Content Understanding, Phi models, Azure SRE Agent.

Agent frameworks & protocols: Microsoft Agent Framework (MAF), Semantic Kernel, AutoGen, LangGraph, LangChain, LlamaIndex, Haystack, CrewAI, DSPy, OpenAI Agents SDK, Google ADK, Mastra, Superagent, AutoGPT, Camel AI; Model Context Protocol (MCP) & MCP servers, Agent2Agent (A2A), multi-agent orchestration, agent lifecycle management, LLM-as-a-judge, human-in-the-loop, tool calling, agent evaluation & observability.

RAG, vector search & retrieval: RAG (embeddings, re-ranking, evaluation), hybrid & semantic search; Pinecone, Weaviate, Milvus, Qdrant, Chroma, FAISS, LanceDB, Vespa, pgvector, Redis Vector, MongoDB Atlas Vector Search, Elasticsearch/OpenSearch Vector, TiDB, Azure AI Search, Amazon OpenSearch Service, Supabase pgvector.

MLOps, GenAIOps & serving: Azure Machine Learning, Azure ML Studio, ML Pipelines, MLflow, Kubeflow, Weights & Biases, DVC, ClearML, BentoML, ZenML, Metaflow, Flyte, TrueFoundry, Feast, Airflow, TFX, Kedro; Model Registry, Feature Store, experiment tracking, AutoML, hyperparameter tuning, GPU orchestration; vLLM, NVIDIA Triton, KServe, Ray Serve, Ollama, llama.cpp, TensorRT-LLM, TGI, SGLang, TensorFlow Serving, TorchServe, Together AI, Fireworks AI, Modal; PyTorch, TensorFlow, JAX, Keras, Hugging Face Transformers, Ray.

Observability, gateways & enterprise AI: Langfuse, Arize Phoenix, LangSmith, Helicone, Weave, Opik, Braintrust, SigNoz, OpenTelemetry, OpenInference; LiteLLM, Portkey, OpenRouter, Cloudflare AI Gateway, Nexus AI Gateway; Databricks, Dataiku, DataRobot, H2O.ai, Domino Data Lab, Alteryx.

Responsible AI & governance: NIST AI RMF, ISO 42001, EU AI Act, Microsoft Responsible AI Standard, Foundry Control Plane, Agent 365; content filters, safety evaluations, adversarial ML, prompt-injection / data-poisoning / model-extraction prevention, AI inventory, risk tiering, data lineage, model watermarking, data provenance, Constitutional AI.

Data & Analytics

Stack: Microsoft Fabric & OneLake, Power BI, Azure Synapse, Databricks (Delta Lake, Unity Catalog), SQL Server (billion-row), Cosmos DB, Azure Data Factory, Data Explorer, Stream Analytics, Event Hubs/Grid; AWS Glue, EMR, Kinesis, MSK, OpenSearch, Lake Formation, QuickSight, Redshift; GCP BigQuery, BigQuery ML, Dataflow, Dataproc, Pub/Sub; Snowflake; T-SQL, KQL, DAX, PySpark, Spark SQL, PL/SQL; medallion & lakehouse architecture, ETL/ELT, data governance.

Cybersecurity & Zero Trust

Stack: Zero Trust (NIST 800-207, CISA); SIEM/SOAR (Microsoft Sentinel, Splunk ES, IBM QRadar, Cortex XSOAR); XDR/EDR/MDR (Defender XDR, CrowdStrike Falcon, SentinelOne, Darktrace); IAM/PAM (CyberArk, BeyondTrust, Delinea, Entra ID Governance); CSPM/DSPM/CIEM, CNAPP, ASM/BAS; DLP/CASB/SASE (Zscaler, Netskope, Prisma Access); Microsoft Purview & Security Copilot; NGFW (Palo Alto, Fortinet, Check Point, Cisco); microsegmentation (NSX-T, Illumio); passwordless & Conditional Access (FIDO2, Windows Hello, PIM/JIT); MITRE ATT&CK/D3FEND; incident response & digital forensics; IoT/OT/SCADA/ICS security; SAST/DAST/IAST/RASP; Confidential Computing/TEE, homomorphic encryption, post-quantum cryptography, zero-knowledge proofs.

Infrastructure, Virtualization & Storage

Stack: VMware vSphere/ESXi/vCenter/NSX-T/vSAN/VCF, Hyper-V & SCVMM, Nutanix AHV/Prism, Proxmox/KVM, Citrix; HCI; Windows Server (2000–2025), Active Directory (multi-forest), Linux (RHEL, Ubuntu, SUSE, CentOS, Debian, Rocky, Alma), UNIX (AIX, Solaris, HP-UX, FreeBSD); SAN/NAS/iSCSI (NetApp, Dell EMC, HPE, Pure

Storage), Fibre Channel/FCoE, Storage Spaces Direct, Azure Local (Azure Stack HCI), Azure VMware Solution, Azure Stack Edge; datacenter & colocation operations; Ceph, GlusterFS, MinIO.

DevOps, SRE & Platform Engineering

Stack: Terraform, Bicep, ARM, CloudFormation, Pulumi, Crossplane, AWS CDK, OpenTofu, Terragrunt; Ansible, Puppet, Chef, SaltStack, PowerShell DSC; GitHub Actions, GitLab CI/CD, Azure Pipelines, Jenkins, CircleCI, ArgoCD, Flux, Tekton, Spinnaker, Harness; Kubernetes (AKS/EKS/GKE/OpenShift/Rancher/K3s), Docker, Helm; Istio, Linkerd, Consul, Envoy; Backstage, Port, Humanitec, Cortex, KubeVela, Dapr; Gremlin, Chaos Toolkit, Litmus, Azure Chaos Studio, Chaos Mesh; Prometheus, Grafana, Datadog, Dynatrace, New Relic, ELK, OpenTelemetry, Jaeger, Tempo, Loki, eBPF/Cilium, Falco; SLO/SLI/SLA & error budgets, Policy-as-Code (OPA/Rego), DevEx.

Programming, Frameworks & Databases

Languages: Python, JavaScript, TypeScript, Java, C, C++, C#, Go, Rust, PHP, Ruby, Perl, PowerShell, Bash, SQL, T-SQL, PL/SQL, KQL, DAX, PySpark, R, Scala, Kotlin, Swift; plus legacy Assembly, COBOL, FORTRAN, Visual Basic, Delphi, VBScript, VBA, FoxPro, Clipper, ColdFusion, Classic ASP.

Frameworks: .NET / ASP.NET Core, Blazor, MAUI, gRPC, Web API; Spring (Boot/Cloud/Security), Quarkus, Micronaut; Django, Flask, FastAPI, Celery; Laravel, Symfony, WordPress, Drupal, Magento; React, Angular, Vue, Svelte, Next.js, Node/Express, NestJS, GraphQL, Prisma; React Native, Flutter.

Databases: Azure SQL/Managed Instance, Amazon RDS/Aurora, Cloud SQL/Spanner, Oracle (Autonomous/Exadata/RAC), IBM DB2, SQL Server, PostgreSQL, MySQL, MariaDB, CockroachDB, TiDB, YugabyteDB; Cosmos DB, DynamoDB, Firestore, Bigtable, MongoDB, Cassandra, Couchbase, Neo4j, Redis, ScyllaDB; Synapse, Redshift, BigQuery, Snowflake, Databricks, ClickHouse, Druid, StarRocks.

Mail, DNS, Web, Messaging & Search

Stack: Exchange Server & Online, Postfix, Sendmail, Exim, qmail, Dovecot, Cyrus, SpamAssassin, ClamAV, Amavis, Rspamd, OpenDKIM/OpenDMARC, Zimbra, Axigen, MDAemon, Kerio, IceWarp, IBM Domino, GroupWise, Roundcube, Horde; BIND, PowerDNS, Unbound, Knot DNS, NSD, djbdns, CoreDNS; IIS, Apache, NGINX, Caddy, Traefik, Envoy, HAProxy, Tomcat, JBoss, WildFly, WebLogic, WebSphere; Kafka, Pulsar, RabbitMQ, ActiveMQ, NATS, Redpanda; Elasticsearch, OpenSearch, Solr, Vespa.

Networking, Identity & Migration/BCDR

Networking: Cisco/Juniper/Arista, BGP/OSPF/EIGRP/IS-IS/MPLS, Segment Routing/SRv6, VXLAN-EVPN; IPv4/IPv6, IPsec/SSL VPN, WireGuard, GlobalProtect, AnyConnect; SD-WAN (Meraki, VeloCloud, Fortinet, Prisma, Versa, Cato); F5 BIG-IP, NGINX Plus, HAProxy; Cisco ISE, Aruba ClearPass, Forescout; NSX-T, Cisco ACI, Calico, Cilium; 5G/4G core, SONET/SDH, DWDM, Carrier Ethernet; gNMI/NETCONF/RESTCONF/YANG telemetry.

Identity: Active Directory (DS/LDS/CS/FS/RMS), Entra ID (Governance, Connect, Domain Services, Verified ID, External ID, Workload ID); SAML, OAuth 2.0, OIDC, Kerberos, LDAP, RADIUS; FIDO2, WebAuthn, passkeys, PIV/CAC, PIM/JIT/PAM; Okta, Ping Identity, JumpCloud, SailPoint, Keycloak, ForgeRock, Auth0, ADFS.

Migration & BCDR: Azure Migrate/ASR, AWS MGN, GCP Migrate, ADMT, Quest, BitTitan, SkyKick, CloudM, ShareGate, SPMT; Veeam, Commvault, Rubrik, Cohesity, Acronis, Veritas, Druva, Datto; Zerto, Site Recovery Manager; 3-2-1/immutable/WORM, Oracle Data Guard/GoldenGate, SQL Always-On, AWS/Azure DMS, Debezium, Qlik Replicate.

ITSM, Project Delivery & Collaboration

Tools & methods: ServiceNow, Jira Service Management, BMC Helix, ManageEngine, Freshservice, Zendesk; CMDB (Device42, NetBox, Lansweeper, IT Glue); Jira, Confluence, Asana, Monday.com, ClickUp, Notion, Linear, Miro, Figma; Scrum, Kanban, SAFe, Lean, Design Thinking; ITIL (change, incident, problem, release, knowledge, service catalogue).

Emerging & Trending Technologies

Focus areas: Agentic AI, physical AI, multimodal AI, domain-specific models, synthetic data & simulation, AI personas, bio-digital interfaces, neuro-symbolic & causal AI, world models, AIOps & self-healing infrastructure; confidential computing/TEE, homomorphic encryption, post-quantum cryptography, zero-knowledge proofs, AI security, model watermarking, decentralized identity & verifiable credentials; blockchain/Web3, smart contracts, digital twins; quantum computing (IBM Quantum, Google Quantum AI, Qiskit, Cirq), neuromorphic computing, silicon photonics, AI chips (Trainium, Inferentia, TPU), CXL, WebAssembly, eBPF, edge AI, 5G/6G; spatial computing/XR, GitOps, FinOps, GreenOps, DevSecOps, platform engineering.

LEADERSHIP, CONTINUOUS LEARNING & R&D

Executive & advisory: C-suite and board advisory, fractional CTO/CIO/CISO, technology strategy & digital transformation, enterprise architecture (TOGAF), cloud adoption strategy, P&L accountability, vendor negotiation, budgets \$5M+, team leadership (50+ staff), organizational design, change management, M&A integration, product roadmap.

Consulting & communication: Stakeholder management, trusted-advisor partnering, pre-sales & solutioning, discovery workshops, proposal/SOW writing, RFP response, cross-cultural delivery (8 countries, 3 continents), executive storytelling, architecture diagramming, technical writing, knowledge transfer & enablement, mentoring & coaching.

Delivery & critical thinking: Systems thinking, problem decomposition, decision-making under ambiguity, risk management, prioritization, ownership & accountability, bias for action, adaptability & resilience, calm under pressure, incident command.

Continuous learning & R&D: Disciplined upskilling from foundational systems to frontier agentic AI — home-lab experimentation, certification discipline, and active community engagement (Microsoft Ignite, RSA Conference, Black Hat, VMware, OpenAI & Anthropic/Claude practitioner cohorts). Current R&D: agentic AI & multi-agent orchestration, prompt engineering, LLM fine-tuning & quantization (LoRA/QLoRA), RAG evaluation, model observability & MLOps, AI safety & responsible AI, post-quantum cryptography, zero-knowledge proofs, edge AI, federated learning, and platform engineering.

CREDENTIALS, ROLES & PORTFOLIO

Certifications & Professional Development

Microsoft: Azure Solutions Architect Expert (AZ-305), Azure Administrator Associate (AZ-104), Security Operations Analyst (SC-200), Identity & Access Administrator (SC-300), Information Protection Administrator (SC-400), Microsoft 365 Enterprise Administrator Expert, AZ-900/MS-900/SC-900, MCSE Cloud Platform & Infrastructure Charter Member (2016), MCSA Windows Server 2012, MCP, AI Skills Fest 2026.

Microsoft AI (2026): AI-901 Azure AI Fundamentals; AI-103 Developing AI Apps and Agents on Azure; AI-300 Operationalizing ML and Generative AI Solutions; AI-500 Multi-Agent AI Solutions Expert (Beta).

AWS: Solutions Architect Associate & Developer Associate (2018); Solutions Architect Professional, DevOps Engineer Professional, Security Specialty, Advanced Networking, Machine Learning, Database Specialty, AI Practitioner, Generative AI Developer (exam-ready).

VMware, Security, Cloud & Data: VCP5/6/6.5-DCV, VMware Cloud Foundation Administrator (training complete); ISO/IEC 27001:2022 Lead Implementer (PECB), CompTIA Security+/Network+/Linux+/Cloud+/A+, ITIL v4, ISC2 Certified in Cybersecurity, CISSP (exam-ready), CEH & CyberArk Defender (training complete); Google Cloud Digital Leader, Professional Cloud Architect & Generative AI Leader (exam-ready), Fortinet NSE 2; Azure Data (DP-300/420/600/700/900); GitHub (GH-100/200/300/500/900); Google Analytics IQ.

Anthropic / Claude (21 credentials, 2026): AI Fluency Framework & Foundations, AI Fluency for Builders, Building with the Claude API, Claude Code in Action & 101, Claude Platform 101, Claude 101, Introduction to Claude Cowork, MCP: Advanced Topics & Introduction to MCP, Introduction to Subagents, Introduction to Agent Skills, Claude with Vertex AI, Claude with Amazon Bedrock, and AI Fluency for education & business — among the earliest cohort of certified Claude / agentic-AI practitioners.

Education

Degrees: Master's, Information Systems Security / Information Assurance — University of the People; Bachelor of Computer Science (BCS) — University of the People; Post Graduate Diploma, Software Engineering — Abaseen Institute (AIMMS) / SDC Punjab (Grade A); Diploma, Software Engineering — AIMMS / Trade Testing Board KPK (Grade B); HSSC, Pre-Engineering.

Awards & Recognition

Highlights: Winner — Microsoft AI Innovation Hackathon (Nov 2025); Charter Member — MCSE: Cloud Platform & Infrastructure (2016); Early Practitioner — among the earliest cohort of certified Claude / agentic-AI practitioners (21 Anthropic credentials).

Roles Delivered & Target Titles

Executive & advisory: CTO, CIO, CISO, Chief Architect, Chief Digital / Innovation Officer, VP / Head of Engineering, Cloud, Platform, Infrastructure, AI, or Security; Fractional / Interim CTO · CIO · CISO; Board & founder technical advisor.

Architecture: Principal / Enterprise / Solutions / Cloud / Systems / Security / Data / AI / Application / Integration / Infrastructure / Platform / Network Architect; Distinguished Engineer; Technical Fellow.

Engineering & delivery: Principal / Staff / Senior Cloud, Infrastructure, Platform, DevOps, DevSecOps, SRE, AI/ML, GenAI, LLM/Agentic AI, MLOps, Data, or Security Engineer; Director / Manager of Engineering, Cloud, Infrastructure, Platform, Security, or IT; Technical Program / Project Manager; Solutions / Presales Consultant; Practice or Delivery Lead.

Scope: Titles apply across seniority (Associate → Senior → Staff → Principal → Lead → Manager → Director → VP → Head → Chief) and industries — banking, insurance, healthcare, government & defense, retail, telecom, SaaS/technology, energy, manufacturing, legal, and logistics — across North America.

Industries Served

Sectors: Retail, Legal, SaaS & Technology, MSP/MSSP, Healthcare, Financial Services, Insurance, Real Estate, Transportation & Logistics, Telecommunications, Media & Entertainment, Hospitality, Pharmaceuticals, Automotive, Aerospace & Defense readiness, Government / Public-Sector readiness.

Key Clients & Partners

Selected (publicly disclosable): Canada — Gore Mutual Insurance, SickKids Hospital, Ottawa Airport, Town of Milton, Lewis Legal / Lewis Associates LLP, Pari Law, Path Law, Allied Immigration, Countax Services, Global Travel & Tours, Minhaj TV Production, MMCC Charity, Healthy Planet, Canada Cloud Solution, Reflect Integrated (MSP), Calians (MSSP), Vet Strategy Health, Zoho Canada, Cronomagic Canada, Hays. Saudi Arabia — Riyadh Bank, Alinma Bank, Tokio Marine Insurance, Alinma Tokio Marine, Hussein Aoueni Co. Bahrain — Arab Eastern Insurance. Japan — Tokio Marine Insurance. Pakistan — Pak/Essa/Attock Cement. India — 3i Infotech. Romania — Axigen. USA — Blue Sprout Agency, GreenVault.

Under NDA (industries only): Government ministries & defense agencies (Middle East), defense-sector technology contractors (Middle East & North America), major regional banks, global insurance companies, healthcare technology partners, and global MSP/MSSP partners.

Delivered technology across: 42+ clients, 8 countries, and 3 continents — North America (Canada, USA), Middle East (Saudi Arabia, Bahrain), and Asia & Europe (Pakistan, Japan, India, Romania).